SKILLS REASONING 6

2

A student carried out some reactions between solutions of chlorine, bromine and iodine, and solutions of potassium halides. Some of their results are shown in the table:

	Potassium chloride solution	Potassium bromide solution	Potassium iodide solution
Chlorine solution		orange solution formed	
Bromine solution			
Iodine solution	no reaction	no reaction	

a Explain why the student did not carry out the experiments shaded in dark grey in the table. **(2)**

SKILLS INTERPRETATION

b Complete the table. **(3)**

SKILLS PROBLEM SOLVING 7

c Write a word equation for the reaction between chlorine and potassium bromide. **(1)**

8

d Complete and balance the ionic equation for the reaction between chlorine and potassium bromide solutions: **(3)**

$Cl_2(aq) + Br^-(aq) \rightarrow$

SKILLS REASONING 9

e Explain in terms of electrons why the reaction between chlorine and potassium bromide solution is described as a redox reaction. **(3)**

f Chlorine can be prepared in the laboratory by the reaction between concentrated hydrochloric acid and potassium manganate(VII). The equation for the reaction is:

$2KMnO_4 + 16HCl \rightarrow 2KCl + 2MnCl_2 + 8H_2O + ...Cl_2$

SKILLS PROBLEM SOLVING 8

i The equation is not balanced. Give the number that should be put in front of Cl_2 in order to balance it. **(1)**

SKILLS EXECUTIVE FUNCTION 5

ii Describe a test for chlorine gas. **(2)**

Test:

Positive result:

iii When chlorine dissolves in water some chloride ions are formed. Describe a chemical test to show that a solution contains chloride ions. **(3)**

Test:

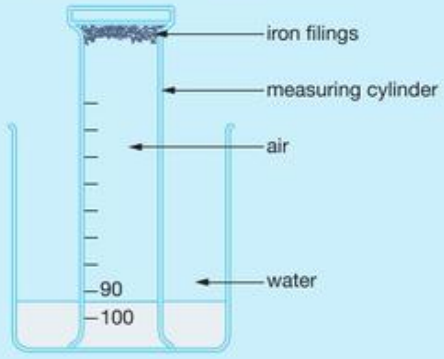
Positive result: **(Total 18 marks)**

SKILLS CRITICAL THINKING

3

a State the name of the most common gas in the air. **(1)**

b A student carries out an experiment to measure the percentage of oxygen in the air. He used the following apparatus:



He carried out the following procedure:

- Put some damp iron filings at the bottom of a 100 cm^3 measuring cylinder.
- Invert the measuring cylinder in a beaker of water.
- Take the initial reading of the level of water in the measuring cylinder.
- Take a reading of the level of water in the measuring cylinder every day for 1 week.

The student's results were:

initial reading on measuring cylinder/cm^3	94
reading on measuring cylinder after 1 day/cm^3	85
reading on measuring cylinder after 2 days/cm^3	80
reading on measuring cylinder after 3 days/cm^3	77
reading on measuring cylinder after 4 days/cm^3	76
reading on measuring cylinder after 5 days/cm^3	75
reading on measuring cylinder after 6 days/cm^3	75
reading on measuring cylinder after 7 days/cm^3	75

SKILLS REASONING 5 6

i Explain why the water level rises. **(2)**

ii Explain why the reading on the measuring cylinder eventually remains constant. **(1)**

SKILLS CRITICAL THINKING

iii Give the name of the iron compound that is formed in this experiment. **(1)**

SKILLS REASONING

iv The student trapped the gas in the tube and put a lighted splint into it. Explain why the splint goes out. **(2)**

SKILLS PROBLEM SOLVING

v Using the results in the table, calculate the percentage oxygen present in the air in the measuring cylinder after 1 day. **(4)**

SKILLS CRITICAL THINKING 7

vi The student decided that 1 week is too long to wait for results so suggested using a more reactive metal such as calcium or lithium in this experiment. Explain one reason, other than that it might be too dangerous, why the experiment will not work with these metals. **(2)**

SKILLS EXECUTIVE FUNCTION 5

vii Suggest one change that the student could make to the experiment so he does not have to wait so long for the results. **(1)**

c Sulfur reacts with oxygen. A teacher burns a piece of sulfur in a gas jar of oxygen.

SKILLS CRITICAL THINKING

i State the colour of the flame. **(1)**

SKILLS PROBLEM SOLVING 9

ii Write a balanced equation for the reaction that occurs. **(1)**

SKILLS REASONING 6

iii A few drops of litmus solution are put into the gas jar at the end of the experiment. State and explain what colour will be seen. **(2)**

d Sodium reacts with oxygen to form sodium oxide.

SKILLS PROBLEM SOLVING 9

i Write a chemical equation for this reaction. **(2)**

SKILLS CRITICAL THINKING 5

ii When sodium oxide reacts with water an alkaline solution is formed. State the name of the ion responsible for making the solution alkaline. **(1)**

(Total 21 marks)

SKILLS DECISION MAKING (6)

4 This question is about copper(II) sulfate solution.

a Describe a procedure to make a solution of copper(II) sulfate starting from copper(II) oxide. **(3)**

(5)

b Describe a chemical test that can be used to show that the solution contains sulfate ions. **(2)**

SKILLS CRITICAL THINKING

c A student added some sodium hydroxide solution to the copper(II) sulfate solution. State the name of the blue precipitate formed. **(1)**

d A student carried out an experiment to investigate displacement reactions of three different metals with copper(II) sulfate solution. She followed this method:

- Put some copper(II) sulfate solution into a polystyrene cup.
- Measure the initial temperature.
- Add some metal.
- Stir the solution.
- Record the maximum temperature.

SKILLS EXECUTIVE FUNCTION

i The student uses the same amount in moles of each metal in the experiments. State two other variables the student should keep the same in this experiment to allow her to be able to compare the different metals. **(2)**

SKILLS ANALYSIS (6)

ii The student recorded the following data.

Metal	Temperature change/°C
zinc	15
silver	0
nickel	7

Arrange the four metals, zinc, silver, copper and nickel, in order of reactivity, starting with the most reactive. **(1)**

SKILLS REASONING (7)

iii The student did not carry out the experiment with copper and copper(II) sulfate. Predict, giving a reason, what the temperature rise would be for this experiment. **(2)**

(8)

iv The student carried out the experiment using 3.25 g of zinc (0.05 mol). She then used the same amount in moles of nickel and silver. Another student decided to use the same mass (3.25 g) of each metal. The results of the two students were very similar. Suggest reasons for this. **(3)**

SKILLS PROBLEM SOLVING

v The ionic equation for the reaction between copper(II) sulfate solution and nickel is:

$$Ni(s) + Cu^{2+}(aq) \rightarrow Ni^{2+}(aq) + Cu(s)$$

Use the ionic equation to explain whether nickel is oxidised or reduced in this reaction. **(2)**

SKILLS EXECUTIVE FUNCTION (6)

e The student wants to make some anhydrous copper(II) sulfate. Describe how she could do this starting with copper(II) sulfate solution. **(2)**

f The student carried out a flame test on the anhydrous copper(II) sulfate.

(5)

i Describe how to carry out a flame test. **(2)**

SKILLS CRITICAL THINKING

ii State the colour of the flame produced when the student carried out a flame test on copper(II) sulfate. **(1)**

(Total 21 marks)

CHEMISTRY ONLY

SKILLS CRITICAL THINKING 4

5

Molybdenite is an ore of molybdenum. The mineral it contains is MoS_2.

The following set of reactions are used to produce pure molybdenum(VI) oxide (MoO_3), from which molybdenum metal can be obtained.

The ore is first roasted in air to form MoO_3. The equation for this reaction is:

$$2MoS_2(s) + 7O_2(g) \rightarrow 2MoO_3(s) + 4SO_2(g) \quad \ldots 1$$

The crude compound is now heated with concentrated ammonia solution:

$$MoO_3(s) + 2NH_3(aq) + H_2O(l) \rightarrow (NH_4)_2MoO_4(aq) \quad \ldots 2$$

Ammonium molybdate is crystallised from the solution and when these crystals are heated pure MoO_3 is formed:

$$(NH_4)_2MoO_4(s) \rightarrow MoO_3(s) + H_2O(l) + 2NH_3(g) \quad \ldots 3$$

The molybdenum can be extracted from MoO_3 by heating with carbon.

a Explain the meaning of the word *ore*. **(1)**

8

b Put ticks in the appropriate boxes in the table to classify each of the reactions 1 and 3 above. Each reaction type may occur once, twice or not at all. **(2)**

	Thermal decomposition	Neutralisation	Redox
Reaction 1			
Reaction 3			

SKILLS PROBLEM SOLVING 9

c Molybdenum(VI) oxide can be heated with carbon to form molybdenum metal. The other product of the reaction is carbon monoxide. Write a chemical equation for this reaction. **(2)**

SKILLS REASONING

d Explain which substance is the *reducing agent* when molybdenum(VI) oxide is heated with carbon. **(2)**

8

e Explain what the reaction in c suggests about the position of molybdenum in the reactivity series. **(2)**

f Molybdenum can also be extracted from MoO_3 by heating with aluminium. Use your knowledge of the reactivity series to give a reason why this method is more expensive than heating with carbon. **(2)**

SKILLS EXECUTIVE FUNCTION 5

g Describe a chemical test you could use to show that the product of reaction 2 contains ammonium ions. **(2)**

(Total 13 marks)

6

6

This question is about lead(II) iodide. This is a bright yellow substance that is insoluble in water. In the past it was used as a pigment, but due to its toxicity this no longer happens. It is now used in the development of new types of solar cell.

a Describe a method a student could use to produce a pure, dry sample of lead(II) iodide. **(5)**

SKILLS EXECUTIVE FUNCTION 7

b The student adds nitric acid followed by silver nitrate solution to their sample of lead(II) iodide. The student stated that the formation of a yellow precipitate showed that it contained iodide ions. Explain why the student's conclusion is incorrect. **(2)**

(Total 7 marks)

END OF CHEMISTRY ONLY

8 **7** This question is about hydrogen chloride and hydrochloric acid.

SKILLS INTERPRETATION

a Hydrochloric acid is made when hydrogen chloride dissolves in water. Draw a dot-and-cross diagram for hydrogen chloride. **(2)**

SKILLS CRITICAL THINKING 7

b When hydrogen chloride dissolves in water an acid–base reaction occurs. The equation for this reaction is:

$$HCl(aq) + H_2O(l) \rightarrow Cl^-(aq) + H_3O^+(aq)$$

Explain why this is classified as an acid–base reaction and identify the acid and the base. **(3)**

CHEMISTRY ONLY

SKILLS INTERPRETATION

c Hydrochloric acid can be used to make chloride salts. Complete the table by stating what you would add to dilute hydrochloric acid to make each of the salts. In each case, state whether you would add it as a solid or in solution and state any other products formed. The first one has been done for you. **(4)**

Reagent	Solid or solution?		Salt	Other product
magnesium	solid	hydrochloric acid	magnesium chloride	hydrogen
copper(II) oxide		hydrochloric acid	copper(II) chloride	
	solution	hydrochloric acid	sodium chloride	water
	solution	hydrochloric acid	silver chloride	nitric acid

END OF CHEMISTRY ONLY

SKILLS CRITICAL THINKING

d A student wants to show that an unknown solution she has been given contains chloride ions. She puts a small amount of the solution in a test-tube and adds dilute nitric acid followed by silver nitrate solution.

5 i Describe what would be seen if the solution contains chloride ions. **(1)**

7 ii Explain why hydrochloric acid cannot be added instead of nitric acid. **(2)**

SKILLS PROBLEM SOLVING 9

e Write an equation, including state symbols, for the reaction that occurs when hydrochloric acid is added to solid copper(II) carbonate. **(3)**

(Total 15 marks)

SKILLS CRITICAL THINKING (5)

8

This question is about chemical tests.

Samples of a very pale green solution, **X**, were tested as follows:

Test	Observation
A sample of **X** was acidified with dilute nitric acid and silver nitrate solution was added	A white precipitate (**A**) was formed
A small amount of sodium hydroxide solution was added to a sample of solution **X**	A dark green precipitate (**B**) was formed
Chlorine was bubbled through a sample of **X**	The pale green solution turned yellow (solution **C**)
A small amount of sodium hydroxide solution was added to solution **C**	An orange-brown precipitate (**D**) was formed

a Use the results from the first two tests to identify **X**. **(2)**

(6) b State the name and formula of **B**. **(2)**

c State the name and formula of **D**. **(2)**

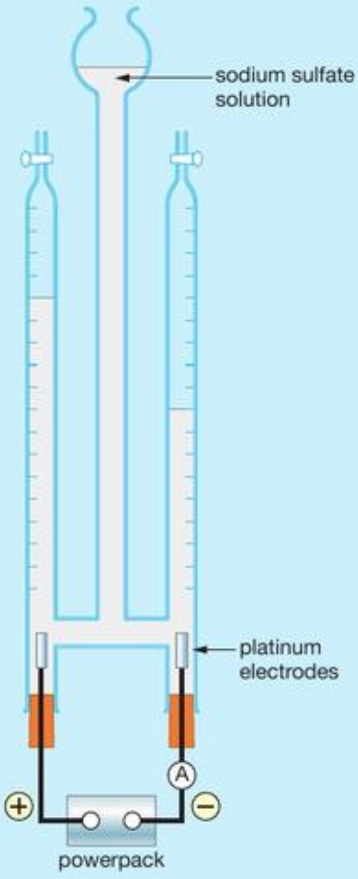
(7) d Explain what type of reaction occurred when chlorine was bubbled through **X**. **(2)**

(Total 8 marks)

CHEMISTRY ONLY

(6) **9**

SKILLS EXECUTIVE FUNCTION



This question is about sodium sulfate.

a State the formula of sodium sulfate. **(1)**

b Describe a practical method by which a student could make a solution of sodium sulfate starting from an acid and an alkali. You should state the names of any pieces of apparatus used and of all chemicals required. **(5)**

c A student used the apparatus shown in the diagram to investigate the effect of changing the time of electrolysis on the volume of oxygen produced when the solution of sodium sulfate was electrolysed.

The student followed this method:

- Set the current to be 0.50 A.
- Record the initial volume of gas in the left-hand arm of the apparatus.
- Leave the current running for 2 minutes and record the final volume of gas in the left-hand arm of the apparatus.
- Repeat for times of 4 minutes, 6 minutes, 8 minutes and 10 minutes.

His results are shown in the table.

Time/min	2	4	6	8	10
Volume of gas/cm^3	3.6	7.2	9.0	14.4	18.0

SKILLS INTERPRETATION 7

i Plot these results on a piece of graph paper and draw a straight line of best fit. **(3)**

SKILLS ANALYSIS, REASONING 4

ii The student suggested that one of the points is anomalous. Identify the anomalous point and suggest the volume of gas that should have been obtained for that time. **(2)**

SKILLS ANALYSIS 5

iii State the relationship between the volume of gas produced and the time of electrolysis. **(2)**

iv Describe a chemical test for the gas produced at the cathode **(2)**

Test:

Positive result:

(Total 15 marks)

END OF CHEMISTRY ONLY

ENERGETICS 207 | RATES OF REACTION 227 | REVERSIBLE REACTIONS AND EQUILIBRIA 240

UNIT 3 PHYSICAL CHEMISTRY

In a world where the population has more than trebled in the last 100 years, fighting against hunger is an extremely important challenge. At the beginning of the 20th century scientists developed industrial processes to make artificial fertilisers that can be used to increase crop yields. In order to develop these processes, chemists and chemical engineers from around the world have had to understand the principles of energy changes, rates of reaction and equilibrium. The world's resources are limited and it is vital that these processes are as efficient and environmentally friendly as possible. This is also the work of scientists.



▲ Figure 19.1 Burning fuel produces enough energy to launch a rocket.